

$0 \notin CI \Rightarrow \text{reject } H_0$

$0 \in CI \Rightarrow \text{fail to reject } H_0$

### HW-4

1.  $y_1, y_2, y_3, y_4 \sim Y \rightarrow t/F$

$(0.34, 4.36, -0.93, 1.37)$

$t \rightarrow -\infty < Y < \infty$ , both +ve & -ve is possible

$F = \frac{U_1/v_1}{U_2/v_2}$  ratio of 2 non-negative quantities

hence  $F \geq 0$

but  $y_3$  is negative so distribution is  $t$ .  
&  $F$  distributed RV can't take -ve values.

2. Power =  $(1 - \beta)$

$\beta \Rightarrow (\text{fail to reject } H_0 \mid H_0 \text{ is false})$

Power  $\Rightarrow (\text{reject } H_0 \mid H_0 \text{ is false})$

$\hookrightarrow$  (c) if we repeat the same test on diff. datasets, the power of the test is the proportion of times you reject a false  $H_0$ .

3.  $H_0: \mu_1 = \mu_2$ ,  $H_1: \mu_1 \neq \mu_2$  two-sided test

$$\mu_1 - \mu_2 = 0$$

$[0.0037, 0.0963] \rightarrow 90\% \alpha = 0.1 \Rightarrow \text{no } 0$

reject  $H_0 \Rightarrow p < 0.1$

$[-0.0051, 0.1051] \rightarrow 95\% \alpha = 0.05 \Rightarrow \text{yes } 0$

fail to reject  $H_0 \Rightarrow p > 0.05$

(b)  $0.05 < p\text{-value} < 0.1$

4. confidence  $\Rightarrow \mu_1 - \mu_2$

we are confident that the procedure prod. intervals containing the true diff. in

the popula<sup>n</sup> means in  $(1-\alpha)$  of repeated samples.

$$\mu = 65$$

$$H_1: \mu > 65$$

$$n = 40$$

$$\bar{X} = 69$$

$$\text{Power} = 0.82$$

$$\alpha = 0.05$$

$$\text{Power} = 1 - \beta$$

$$\beta = 1 - 0.82 = 0.18 \text{ (type II error)}$$

so, 18% of the time they will make a type II error

$$p < 0.05$$

$$p \leq \alpha \Rightarrow \text{reject } H_0$$

that means we reject  $H_0$  82% of the time when the true mean is 69.

so, (b) & (d) are correct.

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\text{width} = 2 z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

so, width gets affected by  $\hat{p}$  (sample proportion),  $n$  (sample size), &  $z_{\alpha/2}$  ( $1-\alpha$ ) (confidence coefficient).